Week 4 Exercises

Note: These exercises are tending toward "tedious" and "rote." However, I want to make sure that we understand the implementation of the basic framework, so there's a lot of tedious repetition in this homework.

Exercise 1 Berk (2010)

Read Berk (2010, journal).

1. Berk distinguishes Level I (descriptive), Level II (statistical inference), and Level III (causal inference) regression analyses. Summarize the features of each level in your own words. Explain why Berk argues that most regressions in criminology are Level I. Does his claim about criminology apply to political science as well?
2. Select a recent empirical article in your subfield that uses non-experimental (i.e., observational) data. Identify how the authors use regression. At what level does the authors *claim* to operate? Use direct quotes if possible. At what level does the analysis *actually* operate? Explain.
3. Berk critiques observational regression treated as causal. He also lists common responses (e.g., "the assumptions are reasonable," "we all have models," "you have to make assumptions to make progress"). Pick one response and assess whether you find it persuasive. Defend your position. How might you design a study in your area that could move from Level I or II toward Level III without weak rhetorical defenses? Perhaps give an example.
4. Berk concludes that, with rare exceptions, regression analyses of observational data should be treated as Level I. Do you agree? If you take Berk's view seriously, what happens to publication standards, peer review, funding, and training?

Exercise 2 Clark and Golder (2006), normal distribution

Clark and Golder (2006) fit the following regression model (that we've seen several times now).

Table 2
The Strategic Modifying Effect of Electoral Laws

	Dependent Variable: Effective Number of Electoral Parties					
	Cross-Sectional Analysis				Pooled Analysis	
Regressor	1980s Amorim Neto & Cox Data[a]	1980s Amorim Neto & Cox Data[a]	1990s Whole Sample	1990s Established Democracies[b]	1946 to 2000 Whole Sample	1946 to 2000 Established Democracies[b]
Ethnic		–0.05 (0.28)	0.06 (0.37)	–0.70 (0.68)	0.19 (0.13)	0.11 (0.14)
ln(Magnitude)		–0.08 (0.30)	0.51 (0.44)	–0.61 (0.59)	0.33* (0.20)	0.08 (0.23)
UppertierSeats	0.04** (0.01)	–0.07 (0.04)	0.01 (0.02)	–0.02 (0.06)	0.05*** (0.02)	–0.06* (0.03)
PresidentCandidates		0.22 (0.27)	0.36 (0.26)	0.07 (0.22)	0.35** (0.16)	0.26* (0.15)
Proximity	–6.05*** (0.88)	–5.88*** (0.84)	–4.19*** (1.26)	–4.95*** (1.24)	–3.42*** (0.55)	–3.10*** (0.46)
Ethnic × ln(Magnitude)	0.39*** (0.07)	0.37* (0.20)	–0.09 (0.17)	0.63* (0.34)	0.08 (0.12)	0.26 (0.17)
Ethnic × UppertierSeats		0.07*** (0.02)	–0.005 (0.01)	0.01 (0.04)	–0.02** (0.01)	0.06*** (0.02)
PresidentCandidates × Proximity	2.09*** (0.26)	1.84*** (0.43)	0.99** (0.46)	1.42*** (0.44)	0.80*** (0.23)	0.68*** (0.23)
Constant	2.40*** (0.21)	2.60*** (0.51)	4.08*** (0.95)	5.15*** (1.32)	2.81*** (0.34)	2.92*** (0.35)
Observations	51	51	62	39	555	487
R^2	.71	.77	.29	.48	.30	.40

Note: Standard errors are given in parentheses for cross-sectional models; robust standard errors clustered by country are used for the pooled models.
a. See Amorim Neto and Cox (1997).
b. Established Democracies omits elections from countries that transitioned to democracy after 1989.
$*p < .10$. $**p < .05$. $***p < .01$.

1. Use an R formula and `model.matrix()` to create the required design matrix `X`. Think carefully about the formula needed.
2. Reproduce their estimates with the matrix multiplication $(X^\top X)^{-1} X^\top y$ or `solve(t(X) %*% X) %*% t(X) %*% y`.
3. Reproduce their estimates using a normal linear model with `optim()`. You can easily adapt code like `est_logit()` from the notes, remembering that you must combine all parameters (i.e., the vector β and scalar σ) into a single vector θ to give `optim()`.

The least squares and ML estimates of the coefficients should match; for the normal linear model, the ML estimate is just the least squares solution.

```
# load Clark and Golder's data
cg <- crdata::cg2006  # from my data package
```

Exercise 3 Long (1997) or King (1998)

Read *either* King (1998) pp. 97-115 (section 5 through 5.3) **OR** Long (1997) pp. 34-84 (chapter 3). Write a short comparison to our notes on the logit model. Highlight the differences you find most interesting or important.

Readings are here.

Hint: You might notice differences in their (i) notation, (ii) justifications for the link function, (iii) recommended quantities of interest, or (iv) treatment of the LPM (foil vs. viable approximation). Pick a couple of these contrasts (or others you discover). Explain each clearly in your own terms. Keep it concise and focused; perhaps 300 words or so.

Exercise 4 Probit

The probit model uses the cdf of the normal distribution (e.g., often denoted as $\Phi(\cdot)$; `pnorm()`) in R.

1. Adapt the code from the notes to fit a probit model. Use the same `ZeligData::turnout` data and model specification `vote ~ age + educate + income + race`.
2. Compare your probit coefficients to the logit coefficients from the notes. Notice that multiplying probit coefficients by 1.6 gives you roughly the logit coefficients.
3. Compute the first difference and 90% confidence interval. Compare to the first difference and 90% confidence interval from the notes.

Exercise 5 Russett and Oneal (2001)

This exercise uses data from Russett and Oneal (2001). The model the probability of a dispute in a dyad-year as a function of several predictors. A quick summary of their book is here. For our purposes, we'll focus on `dem.lo`, which is the lower of the two Polity scores in the dyad. For more details on the variables, see `?crdata::ro2001` or Table 1 on p. 277 of Oneal and Russet (1997).

```
# load russett and oneal data from {crdata}
ro <- crdata::ro2001
```

Fit the logit model `dispute ~ allies + lcaprat2 + contiguity + dem.lo + logdstab + power`.

```
# reproduce their simplest logistic regression model p. 314, table A3.1
f <- dispute ~ allies + lcaprat2 +
  contiguity + dem.lo + logdstab + power
fit <- glm(f, family = "binomial", data = ro)

# extract coefficient estimates
beta_hat <- coef(fit)

# extract covariance estimates
v_hat <- vcov(fit)
```

Use `beta_hat` and `v_hat` from above with the invariance property and the delta method (via `numDeriv::grad()` and `for` loops) to compute the following quantities of interest and their 90% confidence interval.

1. Plot the estimated probability (i.e., expected value) of a dispute as `dem.lo` varies from -10 to 10. Fix all other numeric variables at their medians and `alliesNot Allies`, `contiguityNoncontiguous`, and `powerMinor Powers` at 0. How can these probabilities vary nonlinearly if the `dem.lo` variable is included in the linear predictor *linearly*?
2. What is the change in the probability of a dispute (i.e., "first difference"; $\hat{\pi}_{max} - \hat{\pi}_{min}$) when `dem.lo` changes from its *minimum* to its *maximum*, fixing the other variables at the values described above?[1]

Exercise 6 Berry, DeMeritt, and Esarey (2010)

Replicate Figure 4 Berry, DeMeritt, and Esarey (2009) using `glm()` (or `optim()` if you want more practice), the invariance property, and the delta method with numerical gradients using `numDeriv::grad()`. You can find the `scobit.dta` data set here.

Hints

- The `scobit.dta` data has variables that have been multiplied and squared, etc. *Do not use these variables.* Use R formulas to handle these interactions and polynomials.
- Reading down Table 1 on p. 263, the variables you want are: `newvote`, `closing`, `neweduc`, `age`, `south`, and `gov`.
- The eight distinct values (i.e., 1, 2,..., 8) of `neweduc` correspond to the eight categories that Figure 4 presents along the x-axis.
- You must compute a *first difference* changing `closing` from 0 to its mean. However, you must compute this first difference for all eight values of `neweduc` (i.e., Figure 4 shows 8 first differences). You'll need a plan to compute these *eight* first differences compactly. Perhaps use a for-loop (or `purrr::map` if you want a new challenge).

Exercise 7 Holland (2015)

Holland (2015) writes:

> My first hypothesis is that [the number of] enforcement operations [`operations`] drop off with the fraction of poor residents [`lower`] in an electoral district. So district poverty should be a negative and significant predictor of enforcement, but only in politically decentralized cities [Lima and Santiago]. Poverty should have

[1] I usually recommend computing a first difference as a change from the 25th to the 75th percentile, rather than the min-to the max. The min-max comparison can be sensitive to model specification and represent usual, unrealistic scenarios.

> no relationship with enforcement in politically centralized cities [Bogota] once one controls for the number of vendors (p. 362)"

Use `optim()` to fit a *Poisson regression model* and a *negative binomial regression model* to Holland's data for Santiago. Compare the coefficients and standard errors. Make a side-by-side table of coefficient and SE estimates. Compute a relevant quantity of interest. Explain the differences in the estimates across models.

```
# load data
holland <- crdata::holland2015 |>
  filter(city == "santiago")

# formula corresponds to model 1 for each city in holland (2015) table 2
f <- operations ~ lower + vendors + budget + population
```

Exercise 8 King et al. (1990)

The `ZeligData::coalition` data comes from King et al. (1990). The key variable is `duration`, which is the survival times of government coalitions in parliamentary democracies. Model `duration` using the Weibull distribution (*as described in the notes*–make sure to review this carefully!) using the variables `fract` and `numst2`. Compute the first difference and SE as `fract` moves from a low value to a high value for `numst2 == 0`.

- `duration`: The length of time (in months) that a cabinet survives before dissolution.
- `fract`: Fractionalization index from Rae (1971), measuring the number and relative size of parties in parliament. Higher values indicate more parties of smaller average size (i.e., a more fragmented party system).
- `numst2`: Numerical status of the cabinet, coded 1 for majority governments and 0 for minority governments.

Exercise 9 Clark and Golder (2006); *t* distribution

The code below reproduces the coefficient estimates for the normal linear model from Clark and Golder (2006) as shown in an earlier exercise. Re-estimate their model using a location-scale t distribution. Comment on the differences between the coefficient estiamtes. What does the ML estimate $\hat{\nu}$ of the degrees of freedom parameter ν tell us about Clark and Golder's data?

```r
# load packages
library(tinytable)

# load data
cg <- crdata::cg2006

# create design matrix
f <- enep ~ eneg*log(average_magnitude) + eneg*upper_tier + en_pres*proximity

# normal model
fit <- glm(f, data = cg, family = gaussian)
```

For convenience, here is how me might make a quick table using `tinytable::tt()`.

```r
# make a table
data.frame(term = names(coef(fit)),
           beta_hat = coef(fit),
           se = sqrt(diag(vcov(fit)))) |>
  tt(digits = 2, escape = TRUE) |>
  theme_latex(placement = "H")
```

term	beta_hat	se
(Intercept)	2.916	0.176
eneg	0.112	0.071
log(average_magnitude)	0.078	0.116
upper_tier	-0.057	0.02
en_pres	0.264	0.064
proximity	-3.098	0.352
eneg:log(average_magnitude)	0.264	0.067
eneg:upper_tier	0.059	0.014
en_pres:proximity	0.683	0.137

Hints:

- The R function `metRology::dt.scaled(..., log = TRUE)` computes the required log-likelihood for each observation.
- You need to estimate the $\sigma =$`sd` and $\nu =$ `df` parameters, but they are fixed.

- Model the `mean` as a with the usual linear predictor $X_i\beta$. No link function is needed, because `mean` is unbounded.
- When creating the log-likelihood function, the first argument must be a single parameter vector that include all parameters stacked into a single vector. I often call this `theta` and then extract `theta` into `beta`, `sigma`, and `nu` immediately inside the function.

Exercise 10 Connections

We have covered a wide range of tools. Use the list below as a menu and sketch the connections between two or more ideas. Represent the connections in any format you prefer; a diagram with arrows might make sense to you or you might prefer short written explanations. Simply noting that two things are connected is insufficient—most things are related in this course. Focus on *why ideas are connected* or *how they fit together.*

*Hint: Definitely describe the connections between maximum likelihood estimates, Fisher information, the invariance property, and the delta method. Try for 2-3 other collections of 3 to 8 things that fit together as well.

- **Mathematical Tools**: log, first derivative, second derivative, gradient, Hessian, matrix multiplication, pdf/pmf, expected value, variance
- **Engines**: OLS, maximum likelihood, method of moments; also likelihood function, log-likelihood function
- **Distributions**: Bernoulli, exponential, Weibull, log-normal, Poisson, negative binomial, t, normal, beta, uniform
- **Confidence intervals**: parametric bootstrap, Fisher information, delta method, Wald method
- **Hypothesis Tests**: *nothing so far*
- **Quantities of Interest**: invariance property, odds, exponential rate/mean, beta shape parameters to mean and variance, expected value, first difference
- **Estimate Evaluation**: predictive distribution
- **Estimator Evaluation**: sampling distribution, bias, consistency, coverage, asymptotic distribution of estimator, Monte Carlo simulation
- **R**: `optim()`, `numDeriv()::grad()`, `%*%`, density + distribution + quantile + random number generator functions for various distributions
- **Data Sets**: toothpaste cap, `ZeligData::turnout`, Clark and Golder's (2006) `crdata::cg2006`, Lahman's `batting_average`, WDI, `faithful`, Herron's `hockey`, Holland's (2015) `crdata::holland2015`, the UF Election Lab turnout rates, `ZeligData::coalition`

Exercise 11 Be Creative

We know how to build models.

1. Choose a distribution for y. Many choices here.
2. Choose the parameter(s) to model as functions of design matrix X and those to model as fixed. 1-3 options here, usually.
3. Choose an inverse link function that maps the unbounded linear predictor $X_i\beta \in \mathbb{R}$ onto the parameter space of the modeled parameters. A few options here.

Make creative choices at each step; see what model you can build! What sorts of data might your model fit well? (No need to fit or write the exact log-likehood.)

References

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